different color: RGB.
Step 2. For each point, measure the distance to all centroids, and assign the point to the closest one. At the end, each centroid `C` will define a cluster, a sphere centered at `C` and whose radius is the distance to the furthest point assigned to `C`. The clusters are highlighted in the same color as their centroids.
Step 3. Update the centroids: for each cluster, compute its center of mass.
Step 4. Repeat steps 2 and 3 `j` times. Some points will switch to a different cluster.

12.2 K-means

Let's start our discussion with a classic algorithm, one with a history of success going back to the 50s: k-means is a partitioning algorithm that gathers data in a pre-determined number of spherical clusters.

Figure 12.3 illustrates how k-means works: we can break down the algorithm in three high-level steps:

1. **Initialization**: Create k random *centroids*, random points that can or cannot belong to the dataset, and will be the centers of the spherical clusters;
2. **Classification**: For each point in the dataset, compute the distance to each centroid, and assign the aforementioned point to the closest among the centroids;
3. **Recentering**: The points assigned to a centroid form a cluster: for each cluster, compute its center of mass (as